

Biotic and Abiotic Effects of Remnant and Restoration Soils on the Performance of Tallgrass Prairie Species

Sandi Faber and John Markham

ABSTRACT

Interactions between plants and soil microbes are increasingly recognized as an important component in the functioning of ecosystems. Because these interactions affect and are affected by soil abiotic conditions, restoration efforts must consider the interactions between the plant community, the soil community, and the soil abiotic conditions. We sampled soil from 20 independently restored tallgrass prairies and 8 natural prairie remnants in southern Manitoba. Soils from the restored sites had 4.5 times higher phosphate levels than soils from the remnants. In whole soil assays, big bluestem (*Andropogon gerardii*) and Culver's root (*Veronicastrum virginicum*) had significantly greater growth in soil from the remnant sites. A second growth assay using sterile and inoculated soil from a subset of these sites showed that while big bluestem benefited from soil biota on both remnants and restored sites, the effect was twice as strong on the remnant sites. Our results suggest that plants on restored prairies are less reliant on soil microbes due to the higher fertility found within their soils. Our data suggests that like other ecosystems, residual high fertility in tallgrass prairies may facilitate invasion by non-native plants.

Keywords: mycorrhizae, plant/soil interactions, restoration, soil fertility, tallgrass prairie

The degradation of plant communities alters both abiotic and biotic properties of the soil and may render an infertile medium for future plant growth. In extreme cases of soil degradation, a number of modifications to the soil may be required for plants to grow successfully (Bradshaw and Chadwick 1980, Perry et al. 1989). Bradshaw (1998) emphasized 3 principal issues for consideration before plants are reintroduced for community restoration, including: 1) amending the physical habitat (e.g., soil texture, structure, stability, and moisture); 2) amending the chemical components (e.g., macro- and micro-nutrients, pH, heavy metals, and salinity); and 3) removing exotic plant species. While most restoration work examining plant/soil interactions has

focused on soil abiotic properties, the biotic soil community in plant community re-establishment is increasingly being recognized as playing an important role, with soil microbes being critical to overall community health and ecosystem function (Young et al. 2005, Eviner and Hawkes 2008, Heneghan et al. 2008, Harris 2009, Kardol et al. 2010). Examples include the successful restoration of salt marsh ecosystems in the eastern United States where native grasses were inoculated with arbuscular mycorrhizal fungi (AMF) to establish species cover and stabilize salinity levels (McHugh and Dighton 2004), clear-cut forests along the northwestern coast of the United States where soil inoculation from uncut forest was used to improve tree growth and survival (Perry et al. 1989), pre-inoculation of planted grasses with AMF to increase growth and survival in abandoned agricultural fields in Arizona (Richer and Stutz 2002), and the use of soil pathogens to

control invasive plant species (Kremer et al. 2006).

While in certain restoration sites, soil amendments must be added to increase plant survival, areas that have been used for agricultural production may have higher levels of soil nutrients than natural systems, albeit with lower levels of variability in their distribution (Baer et al. 2003). Both high nutrient levels and lower heterogeneity can reduce diversity in plant communities (Tilman 1984, Inouye and Tilman 1995, Thompson et al. 2005). Additionally, interactions between plants and soil biota are known to depend on soil nutrient levels (DeDeyn et al. 2004). In particular, increased levels of soil nutrients can decrease plant reliance on mycorrhizal interactions. When non-native species are less reliant on these interactions their invasiveness can be increased (Eviner and Hawkes 2008). Therefore, restoration efforts must consider the environmental

conditions under which plant/soil microbe interactions will occur.

The rarity of tallgrass prairies have made them a special concern for ecological restoration (Kline and Howell 1987, Richter and Stutz 2002, Martin et al. 2005, Polley et al. 2005). Studies have found a positive influence of biotic soil components, such as arbuscular mycorrhizae, on establishment of native species (Smith et al. 1998). This confirms the more general ecological studies showing the dependence of native tallgrass species on mycorrhizae (Hetrick et al. 1989, Hetrick and Wilson 1991, Hartnett and Wilson 1999). While the primary measure of restoration success is the composition of the macro flora and fauna (Allison 2002), there is increasing awareness of the importance re-establishing plant/microbe interactions and the effect plant and soil communities and their interactions have on ecosystem processes (Eviner and Hawkes 2008). It is unclear if restored grassland communities interact with the soil community in the same way as remnant communities do. The goals of this study were to compare the effects of soil from restored tallgrass prairies to soil from prairie remnants on the performance of native species. We also determined if these effects were due to the abiotic or biotic properties of the soil. This will allow us determine which, if any of these properties is more important in establishing plant/soil interactions during tallgrass prairie restoration.

Methods

Study Area

We selected 28 sites in the Red River Valley of southern Manitoba for soil sampling in spring 2006 (Appendix 1). We classified 8 sites as remnant prairies, defined as areas that have never been cultivated, which represent most of the total area of tallgrass prairie in Manitoba. The remaining 20 sites were classified as restored prairies, defined as areas previously used for

either growing crops or urban development and subsequently planted with native tallgrass prairie species. The average cover of exotic species in the restoration sites was 13% (with a maximum of 36%), whereas the remnants had an average exotic species cover of 3% (with a maximum of 6% (Paul Mutch, personal communication). For more information on the sites, see Mutch (2008).

Soil Assays

To evaluate plant/soil interactions, we conducted 2 greenhouse growth assays. As test species we used big bluestem (*Andropogon gerardii*) and Culver's root (*Veronicastrum virginicum*). Big bluestem was chosen because it is the most indicative plant of tallgrass prairies and is known to have a high mycorrhizal dependence, even compared to other C4 grasses (McCain et al. 2011). Culver's root is rare in tallgrass prairies and in our region is only found in the most unaltered sites. We expect it, like many forbs, to be less dependent on mycorrhizal interaction (Wilson and Hartnett 1998). We used local seed sources from prairie remnants since it has been shown that seedlings from these sources can interact differently with prairie soils than seedlings from non-local, or bred plant stock (Gustafson et al. 2004).

We first ran a whole soil assay to compare plant performance in soil from prairie remnants and restored sites. We chose to use undisturbed soil cores for the growth assay to include the possible effects of fungal networks on plant growth (Johnson et al. 2003). Using a bulb planter, we collected individual cores (7.3 cm diameter by 10 cm deep) from the sites. We sampled 10 1-m × 1-m plots evenly spaced across the sites and collected 2 cores per plot (20 cores per site) for a total of 560 soil cores. We wrapped the cores in polyethylene bags and placed them in cold storage (4°C) for a period of up to 3 wks between collection and planting to ensure the soil microbes were active. We collected seeds of big bluestem from a tallgrass

community at the Oak Hammock Marsh site (Appendix 1) in 2005. The seed was air-dried and stored at -20°C prior to seeding. We purchased native seeds of Culver's root from a local seed company (Prairie Habitats, Inc., Argyle, MB), and we confirmed that the parent plants came from locally collected seed originally harvested from a number of prairie remnants. We cold-stratified these seeds at 4°C in moistened potting mix for 4 wks prior to seeding. We sowed the seed of both species on trays of sterilized Turface™ (Profile Products, Buffalo Grove, IL) in a greenhouse, with no fertilizer added, and allowed them to establish for 2 wks before planting.

We set up our experimental pots (10 cm diameter by 9 cm depth) in mid-July 2006 with one soil core positioned top-up in each pot. To maintain the shape of the plastic wrapped core, we packed the surrounding space in the pot with perlite. We then cut back the polyethylene covering on each core to expose the soil surface and punctured holes underneath to allow for drainage. We transplanted one seedling of big bluestem or Culver's root into a core from each plot and arranged the pots in a completely randomized design on greenhouse tables for each species. We weeded pots for the first few weeks and watered each through a drip line system for 45 s every other day to keep the soil moist. The plants were maintained under natural light conditions, average growing season temperatures (ca. 24°C daytime/18°C night-time), and no fertilization for 6 wks.

Immediately following harvest, we collected approximately 5 mL of fresh soil from each soil core and stored it at -80°C for use in the inoculation assay. We then measured the total plant fresh mass in each of the cores. In half of the replicates from each site, we stored half the roots of each pot at -80°C for future mycorrhizal examination, and in the remaining replicates, we dried the plants at 65°C. We then determined the root dry mass of plants for which a root subsample

had been collected by using the ratio of dry to fresh mass of the roots, that were dried, multiplied by the total root fresh mass of the plants. We also used the dry to fresh weight ratio at harvest multiplied by the fresh mass at the time of planting to estimate the dry mass at the time of planting. We justified this procedure on 2 grounds. First, an ANCOVA showed no site effect the fresh to dry mass ratio. Second, there was a strong correlation between fresh and dry mass ($r = 0.84$).

For mycorrhizae analysis, we cut root samples into small lengths (~1 cm), cleared them with 2.0 M KOH, acidified them in 1% HCl, and stained them in Trypan Blue (Johnson et al. 1999). We then examined root pieces under a light microscope for percent arbuscular mycorrhizal colonization at 40x magnification using the grid-intersect method (McConigle 1990). Grid intersections were considered colonized when they were overlaid vesicles, arbuscules, or hyphae that could be traced to either of these.

Immediately following the harvest of plants, we selected soil samples randomly from half of the replicates of each site for analyses for gravimetric water content (GWC). The difference between fresh and oven dried at 105°C mass was used to calculate GWC (Jarrell et al. 1999). We determined organic matter from loss on ignition at 500°C for 4–5 hr, from one replicate on each site (Harmon and Lajtha 1999). We measured soil phosphate levels on air dried soils using bicarbonate extracts and the Murphy Riley method (Kalra and Maynard 1991), and soil inorganic nitrogen on air dried samples using the microdiffusion method (Mulvaney 1996) on 5 replicates per site.

To separate the contribution of soil abiotic and biotic properties on plant performance, we also performed a soil inoculation assay; however, due to space limitations, not all sites were assessed. We chose 5 restored and 5 remnant sites at random and planted big bluestem seedlings in soil from

each site. Following harvest we sieved, bulked by site, air-dried, and autoclaved at 121°C for 1 hr, the soil cores from the whole soil core assay. Our past work has shown that this is effective in sterilizing the soil, but fungi present in the greenhouse may colonize plants grown there. We loosely packed the soil in Ray Leach “Cone-tainers”™ (2.5 cm × 12 cm, Stuewe & Sons, Inc., Corvallis, OR). To separate the abiotic and biotic effects of the soil, we grew plants in both pots that were inoculated with the 5 mL of live soil stored at -80°C, and pots with no inoculation treatment (control). The purpose of the sterile soil was to create a background soil of the abiotic conditions for each soil site. The inoculation treatment was applied to incorporate biotic conditions. We placed the live soil in a small hole in the soil next to the planted seedling. We replicated each site by inoculation treatment 5 times. The Cone-tainers were placed in racks (25 per rack), arranged randomly on greenhouse tables, and watered by an overhead-mist system for 2 min every other day to keep the soil moist. We used the same growth conditions as in the whole soil core assay for 8 wks, and we processed the plants in the same manner as the whole soil assay.

Data Analysis

For both assays we calculated the relative growth rate (RGR) for each plant as the difference in log dry mass between the end and start of the experiments, divided by the number of growing days (Hunt 1978). We used ANOVA to compare differences between the 8 remnant and 20 restoration sites in the whole soil core assay. Because each soil core was considered a subsample and the site was considered a true replicate, we used the mean RGR values from the 10 soil cores for each species on each site in the analysis. We compared the RGR of both species using a least squares regression of mean values per site to determine if both species responded in the same manner to the different site

conditions. We also examined relationships between RGR and site variables (age area, soil N, P, pH, organic matter, and gravimetric water content) using least squares correlation. When needed, we log transformed the data to homogenize variances. We used ANCOVA to determine if the relationship between plant growth and environmental variables differed between restored and remnant prairies first by crossing the site type with the environmental variable, and if not significant, running the ANCOVA model again without an interaction term.

For the inoculation assay we examined the effect of soil inoculation on RGR for each site and used a two-way ANOVA followed by a Tukey’s HSD test to examine the effects of site and soil inoculation on plant growth. We evaluated soil phosphate differences between the sites using one-way ANOVAs. Because plant growth differed between sites, we estimated the soil biotic effect (BE) as the relative increase in growth for inoculated plants on each site:

$$BE = \frac{RGR_i - RGR_s}{RGR_i} \times 100\%$$

where RGR_i and RGR_s are the relative growth rates of plants grown in inoculated and sterile soil, respectively. We examined the relationship between mycorrhizal colonization, growth, and site variables using least squares correlation, and we explored differences in root colonization using one-way ANOVA between remnant and restoration sites.

Results

Whole Soil Core Assay

Both big bluestem and Culver’s root had significantly greater RGR on soils from restored, compared to remnant sites, with an increase in RGR of 32% and 38%, respectively (Table 1). Mean growth per site of big bluestem was significantly correlated with growth

of Culver's root (correlation coefficient = 0.77; $p < 0.0001$), indicating that individual sites had the same effect on plant performance regardless of the species. The size of the sites ranged from 0.07 to 3530 ha (Appendix 1), and the average age of the restoration sites was 8.3 ± 1.2 yr (mean $\pm$ standard error) and ranged from 3–17 yr. Remnant and restoration sites only differed significantly in soil P, GWC (measured at the end of the whole core assays) and size of the site. Phosphate levels were almost 5 times higher on restoration sites, while GWC was 32% lower. The only significant relationships between site variables were between GWC and organic matter (correlation coefficient = 0.38, $p = 0.041$), and GWC and soil inorganic N (correlation coefficient = 0.48, $p = 0.01$).

The growth of big bluestem and Culver's root were both positively correlated with mean soil P per site (Figure 1). The growth of Culver's root was also positively correlated with soil inorganic N levels. None of the other site variables were correlated with the growth of either species and the ANCOVA indicated that the relationship between the growth of both species and soil P did not differ between remnant and restored sites. The overall rate of mycorrhizal fungi infection was $53 \pm 4\%$ for big bluestem, and $24 \pm 3\%$ for Culver's root. There was no difference in rates of infection for either big bluestem grown on remnant versus restored sites ($p = 0.72$) or for Culver's root ($p = 0.89$). There were no significant correlations between rate of growth and mycorrhizal infection for either big bluestem ($r = 0.025$, $p = 0.59$) or Culver's root ($r = 0.17$, $p = 0.53$). Also, there were no correlations between colonization rate and environmental parameters. However, colonization was always less than 40% in big bluestem when soil P was greater than 12 ppm (this occurred on 3 restoration sites). Also, for the 15 sites where the age was known, sites less than 5 yr old always had a big bluestem colonization rate of less than 40%, whereas sites restored more than

Table 1. Differences in plant growth, soil properties, and sizes of restored and remnant prairies, Manitoba. Values are means of sites means ($\pm$ standard error) with p values for one-way ANOVAs. Significant p values are in bold. RGR: relative growth rate, GWC: gravimetric water content, OM: organic matter.

	Remnants	Restorations	p
RGR (mg per g per d)			
big bluestem	41 $\pm$ 4	54 $\pm$ 2	0.001
Culver's root	64 $\pm$ 6	88 $\pm$ 3	0.015
GWC (%)	51.1 $\pm$ 5.0	34.9 $\pm$ 1.87	0.001
OM (%)	18.3 $\pm$ 2.0	18.1 $\pm$ 1.6	0.955
Phosphate (ppm)	1.8 $\pm$ 0.2	8.3 $\pm$ 1.0	0.000
Inorganic N (ppm)	23.6 $\pm$ 3.5	21.2 $\pm$ 1.5	0.459
pH	7.74 $\pm$ 0.13	7.78 $\pm$ 0.04	0.667
Size (ha)	47.0 $\pm$ 43.7	0.234 $\pm$ 0.084	0.006*

*test based on log transformed data

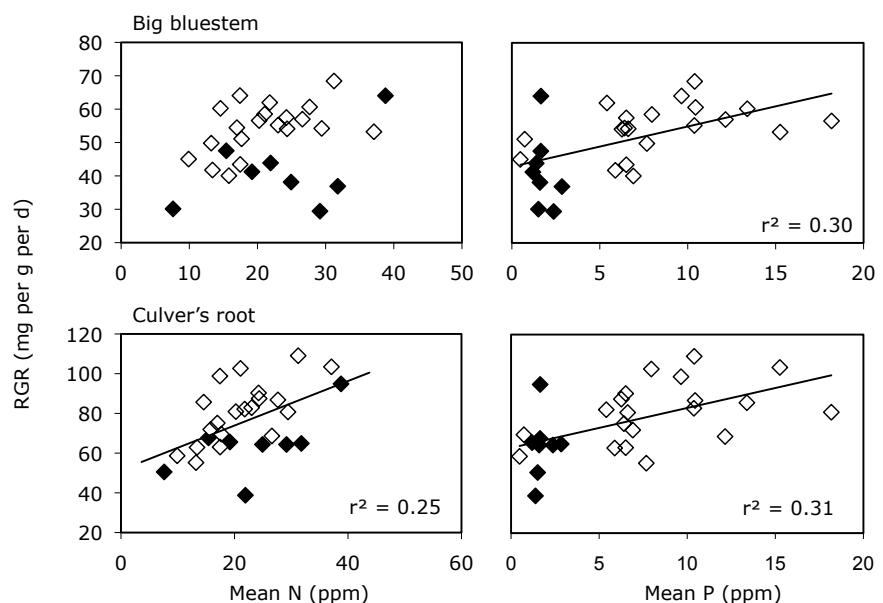


Figure 1. Relationship between plant relative growth rate and soil phosphorus and inorganic nitrogen level (in parts per million) in the whole soil assay. Each point is the mean of the 10 replicates on each site. Open symbols are restored, and closed symbols are remnant sites. Significant correlations are shown.

7 yr ago always had a colonization rate greater than 53% (Figure 2).

Inoculation Assay

As with the whole soil core assay, plants grown on restoration soils had significantly higher relative growth rates (54.2 ± 3.6 mg per g per d) than plants grown on soil from remnants (33.1 ± 4.2 mg per g per d, $p < 0.001$, Figure 3). Plants grown on inoculated soil had significantly greater growth (50.1 ± 4.3 mg per g per d) than plants grown on sterilized soil (37.2 ± 5.3 mg

per g per d, $p = 0.018$). While there was no significant interaction between site type and soil inoculation, when analyzed as a whole, individual sites reacted differently to the inoculation treatment. Only 2 of the 5 restoration sites (1 and 4) had significantly greater performance when soil was inoculated, whereas all but 1 remnant site (27) had significantly greater growth when inoculated (according to ANOVAs performed on data from individual sites). Also, plants grown on soil from remnant sites had a BE twice as large

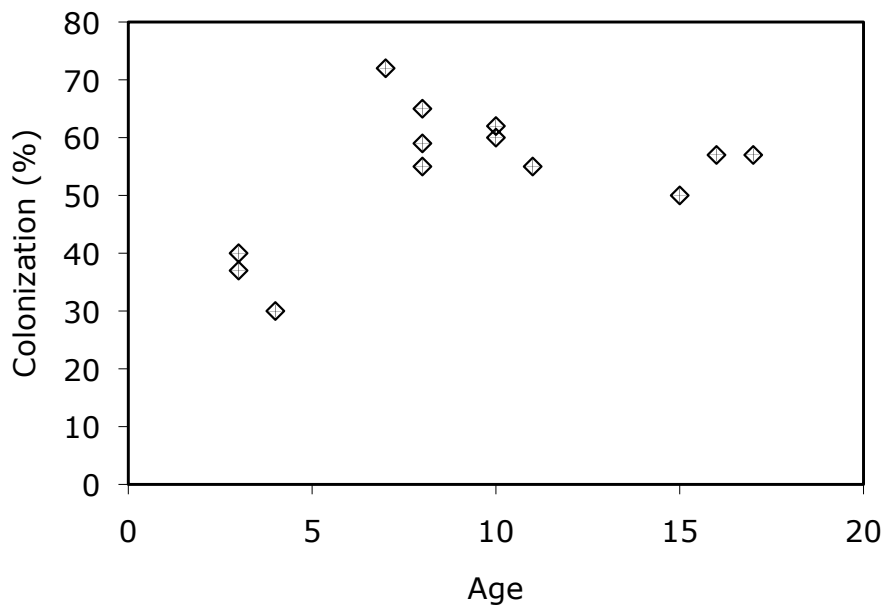


Figure 2. Relationship between big bluestem (*Andropogon gerardii*) mycorrhizal fungi colonization and the time since a site has undergone restoration. Each point is the average of 5 plants grown on soil cores collected from a different site.

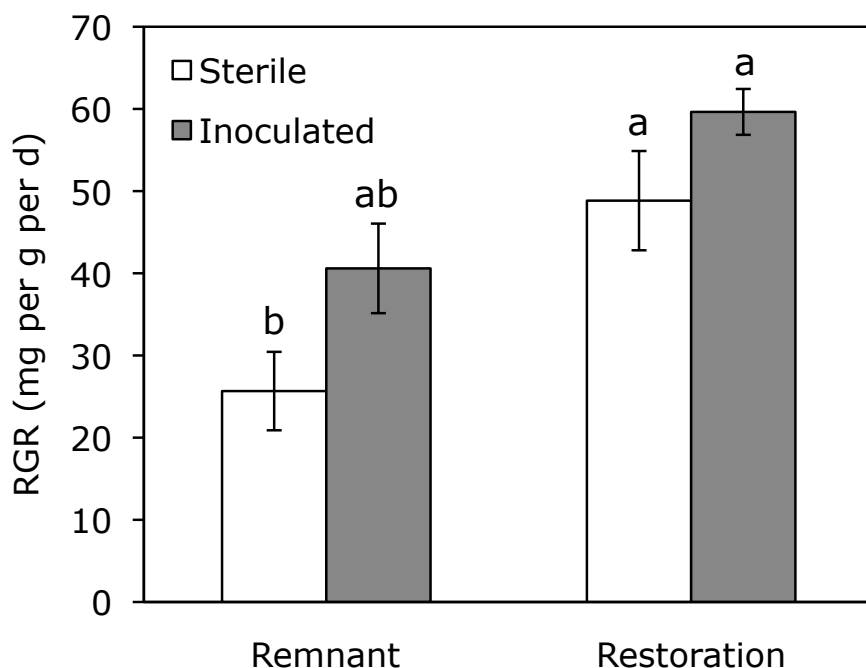


Figure 3. Effect of autoclaved soil from prairie remnant and restored sites, with and without a live soil inoculation, on the growth of big bluestem. Bars are means from 5 sites. Error bars are standard errors. Bars with the same letter are not significantly different according to a Tukey HSD test.

as plants grown on soil from restoration sites ($183 \pm 13\%$ versus $92 \pm 32\%$, $p = 0.030$). There was also a strong negative correlation between BE and the final dry mass of plants (Figure 4), demonstrating that plants benefit

more from soil biota when the soil is less fertile. According to a 2 way ANOVA there was no effect of soil sterilization on soil P ($p = 0.425$), but soils from the remnant sites had lower soil P (2.6 ± 2.0 ppm, $p = 0.019$) than

soil from the restoration sites (9.8 ± 1.4 ppm). There was also a significant positive correlation between the mean site P level and plant final mass ($r = 0.829$, $p = 0.0008$).

Discussion

Our data demonstrate that plants interact with soil from tallgrass prairie restoration sites in a fundamentally different way than with soil from tallgrass prairie remnants. Soil from restored sites was more phosphorus rich, promoting greater growth of both a common and a rare prairie species when grown in intact soil cores. Additionally, plants from prairie remnants were more reliant on the soil biota for their growth, whereas plants from restored sites tended to be unaffected by the presence of soil microbes in the soil. The higher phosphate level in the restoration sites was likely a result of the past level of human nutrient addition and disturbance on these sites. A lower level of disturbance in the remnant sites likely leads to lower levels of P cycling and availability (Vitousek 2004). Although much of restoration ecology is concerned with increasing productivity on degraded lands, achieving greater productivity than natural ecosystems through human disturbance is not a desirable outcome. In the short term disturbances create conditions both for the establishment of exotic species and weedy species seedlings. In the long term they can alter the competitive balance between native and exotics when increased soil fertility lessens the advantage native species gain through interactions with soil microbes (Eviner and Hawkes 2008). While the restoration sites we studied produced greater plant growth, they had a higher proportion of exotic species. Restoration efforts must often control exotic species that benefit from human induced disturbance regimes (D'Antonio and Myerson 2002, Hobbs and Crammer 2008).

The effect of increasing productivity on decreasing biodiversity is a recent phenomenon in agricultural lands (Hodgson et al. 2005). High phosphorus levels have been shown to affect the re-establishment of high diversity grasslands in dunes and chalk grasslands (Hobbs and Huenneke 1992), and management practices in Europe now attempt to address the negative impacts of increased soil fertility on plant diversity (Smith et al. 2003). While our restoration sites span an age range of 14 yr, we saw no indication that soil nutrient levels were decreasing with time. This is not surprising since these restoration sites were independently established and maintained. Developing protocols to reduce nutrient availability should be investigated to re-establish natural plant/soil microbe interactions. For instance, the addition of organic matter to soil can dramatically lower soil nitrate levels and decrease the abundance of exotics in restored grasslands (Blumenthal et al. 2003, Baer et al. 2004). This phenomenon may also result in plant/soil microbe interactions more typical of remnant systems.

While soil fertility levels may take many years to decrease, mycorrhizal associations may be re-established on a much shorter time scale. Many grassland species are obligate mycotrophs that require association with arbuscular mycorrhizae to grow to maturity (Wilson and Hartnett 1998, Wilson et al. 2001). On severely degraded soils, mycorrhizal fungi inoculation can aid in plant re-establishment (Kardol et al. 2010). However, we found the effects of mycorrhizal fungi colonization decrease over time, suggesting that inoculation may not be required in most cases. We cannot rule out that types of mycorrhizal fungi differ with the age of a site, as has been found by others using whole soil core plant assays (Ji et al. 2010). In a comparison of a 2, 12, and 17-yr old tallgrass restoration, Anderson (2008) found that an early successional species, Canada wildrye (*Elymus canadensis*), had higher

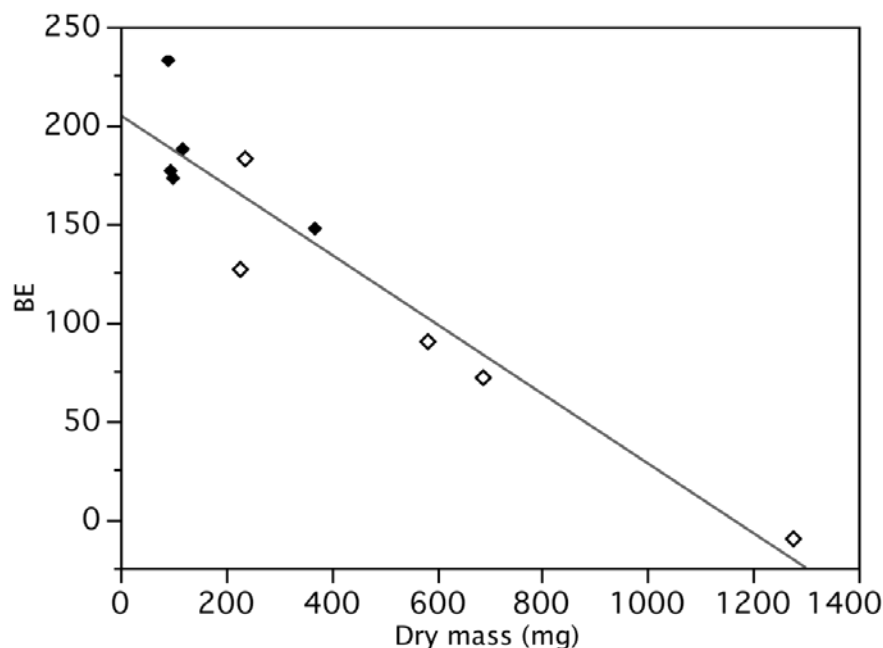


Figure 4. Relationship between the biotic effect of soil (BE) and final dry mass of plants grown on soil from restored and remnant prairies, Manitoba. The line is a least squares fit ($r^2 = 0.90$).

rates of mycorrhizal fungi colonization in the 2-yr old, compared to the 12 or 17-yr old restoration. However, there was no effect of age on the colonization rate or growth in a later successional species, little bluestem (*Schizachyrium scoparium*). White and colleagues (2008) also found that after 1 yr there was no difference in the rate of root colonization on sites they had inoculated with mycorrhizal fungi compared to their control site. Our data agree with other findings that show the rate of mycorrhizal colonization is not necessarily a good predictor of plant growth (Smith et al. 1998, Smith et al. 2004, Busby et al. 2011). We also found little evidence that soil phosphate levels decrease mycorrhizal colonization except on sites with the highest soil phosphate levels. High phosphate levels have traditionally been considered to reduce plant colonization by mycorrhizal fungi and their effectiveness on plant growth (Smith and Read 1997). Our results may be due to the fact that while phosphate levels on our restoration sites were higher than those on remnants, they were still much lower than those found by others where mycorrhizae

are either inhibited or fail to have an effect on plant performance (White et al. 2008). In a recent meta-analysis, Hoeksema and others (2011) also concluded that soil N levels were more important than soil P in determining the effect of mycorrhizae on plant growth.

Although only mycorrhizal fungi were examined in this study, it is important to emphasize that the positive effects of soil biota are likely due to the diversity of microbial species present in the soil community (Bever et al. 1997, Chanway et al. 1991, Bever 2002, Bever 2003, Ehrenfeld et al. 2005, Wolfe and Klironomos 2005, Wardle 2006, Casper and Castelli 2007). Chanway and colleagues (1991) suggest that free-living bacteria (i.e., rhizobacteria) in the soil can enhance plant growth through increasing P solubilisation and N fixation, suppressing antagonistic bacteria, and producing plant growth substances. Nitrifying bacteria have been shown to be associated with the success of exotic plants in California grasslands that preferentially take up nitrate (Hawkes et al. 2005). Given the various effect soil microbes can have on plant

performance, it has been suggested that whole-community interactions within the soil are far too complex to uncouple in assessing effects on the plant community (Wolfe and Klironomos 2005). However, the approach taken here, where the overall effect of soil biota on plant performance is measured, can yield useful insights into plant/soil microbe interactions. Using whole soil communities can also result in stronger mycorrhizal promotion of plant growth (Hoeskema et al. 2011). Overall, our data suggest that restoration efforts should focus more on establishing soil abiotic conditions that would then allow plant/microbe interactions to naturally establish.

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Sandi Faber, Department of Biological Sciences, University of Manitoba, Winnipeg, MB., R2T 2N2, Canada.

John Markham (corresponding author), Department of Biological Sciences, University of Manitoba, 212B Biological Sciences Building, Winnipeg, MB., R2T 2N2, Canada, markhamj@cc.umanitoba.ca.

Appendix 1. Site description of tallgrass prairies studied, Manitoba. Remnant sites (Rm) refer to tallgrass areas not previously disturbed for agriculture or urban development. Restoration sites (Rs) refer to areas disturbed for agriculture or urban development and later planted with native species. P = inorganic phosphate; N = inorganic nitrogen; GWC = gravimetric water content; OM = organic matter. NA = unknown; * = study sites used in the soil inoculation assay.

Site Name	Lat/Long (d:m:s)	Site Type	Age (yr)	Area (ha)	P (ppm)	N (ppm)	GWC (gm/gm)	OM (%)
1* Big Bluestem Park	49:53:12 97:11:44	Rs	8	0.32	5.88	13.40	0.25	14.8
2 Lagimodiere Heritage Park	49:53:47 97:06:52	Rs	NA	0.25	18.18	20.24	0.38	16.6
3* Plessis Bergen park	49:56:58 97:05:30	Rs	NA	1.09	10.39	31.23	0.46	14.5
4* Spence St. park	49:52:53 97:09:16	Rs	7	0.07	9.65	17.43	0.27	12.2
5 Royalwoods prairie buffer	49:49:33 97:04:24	Rs	3	14.6	7.68	13.22	0.37	15.5
6* Harry Collins	49:49:08 97:08:01	Rs	10	0.13	15.26	37.08	0.42	17.2
7 St Andrews	50:03:39 97:00:10	Rs	4	2.70	12.15	26.59	0.40	15.0
8 McBeth	49:57:20 97:04:59	Rs	10	4.05	6.91	15.81	0.22	31.1
9 Living Prairie Museum	49:53:24 97:16:20	Rm	NA	13.6	1.19	19.19	0.50	18.0
10 Sturgeon Creek	49:52:33 97:16:17	Rs	NA	0.27	10.44	27.65	0.31	18.1
11 Charleswood Bridge	49:52:16 97:15:52	Rs	NA	0.17	6.62	29.43	0.34	14.8
12 King's Park	49:47:38 97:07:10	Rs	11	1.20	10.38	23.02	0.34	14.5
13* University of Manitoba	49:48:25 97:07:60	Rs	3	0.09	13.39	14.57	0.24	11.4
14 Smith Carter	49:50:01 97:10:38	Rs	3	3.93	0.48	9.90	0.32	11.7
15 Ferrier prairie	49:57:29 97:07:04	Rm	NA	0.32	1.39	21.93	0.66	15.5
16 Forks	49:53:12 97:07:47	Rs	4	2.21	5.84	24.24	0.55	36.5
17 Manitoba Hydro	49:51:16 97:09:36	Rs	17	8.09	6.51	17.47	0.33	15.6
18 Harbourview	49:56:11 97:01:53	Rs	NA	6.96	0.73	17.67	0.32	13.3
19 Elmwood High School	49:54:39 97:05:52	Rs	15	0.22	6.41	16.98	0.29	18.3

	Site Name	Lat/Long (d:m:s)	Site Type	Age (yr)	Area (ha)	P (ppm)	N (ppm)	GWC (gm/gm)	OM (%)
20	Warsaw	49:51:45 97:10:20	Rs	16	0.13	6.26	24.35	0.37	18.9
21	Murray	49:57:21 97:06:10	Rs	8	0.05	5.40	21.80	0.35	19.1
22	Bay	49:50:55 97:03:54	Rs	8	0.16	7.96	21.06	0.49	33.3
23*	Wilkes prairie	49:50:33 97:15:02	Rm	NA	0.07	1.60	24.93	0.28	10.1
24*	Rotary prairie	49:53:51 97:02:03	Rm	NA	10.0	2.38	29.17	0.45	22.0
25	Plessis prairie	49:52:16 97:01:43	Rm	NA	3.66	1.65	15.42	0.55	26.7
26*	Oak Hammock Marsh prairie	50:10:40 97:09:37	Rm	NA	116	2.85	31.78	0.60	24.0
27*	Tall Grass Prairie Preserve	49:10:42 96:40:26	Rm	NA	3530	1.51	7.61	0.37	11.5
28*	St. Charles Rifle Range	49:54:37 97:20:17	Rm	NA	85	1.65	38.78	0.69	18.5